A Biosynthetic Inverse Compiler for Verifier-Grounded Fungal Genome Mining

The reconstructibility ceiling in fungal polyketide chemistry

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Abstract

Predicting a fungal natural product from its biosynthetic gene cluster (BGC) is usually posed as direct BGC$\to$structure regression; models built this way remain weak (balanced accuracies 51–68%), and pooling a thousand bacterial BGCs with the few hundred fungal ones barely helps. We argue the framing is wrong. A BGC is not a molecular blueprint but a *constrained latent program space*: the product is the image $y = \mathrm{Exec}_\Theta(z; G)$ of a biosynthetic program z under an operator library Θ and a domain-derived grammar G, so genome mining is *inference over* z from physical observables rather than regression. We build a **grammar-agnostic biosynthetic inverse compiler** that realizes this: a grammar-specific, sound deterministic *executor* enumerates legal programs; a shared, adduct-aware, ppm-tolerant *observability layer* collapses the candidate set with accurate mass and MS/MS; and a shared *verifier* returns a three-state reconstructibility verdict—VERIFIED, UNDER-OBSERVED (naming the next observable that would collapse the set), or OUT-OF-GRAMMAR.

Our empirical case study is fungal iterative PKS, where the framing bites hardest: bacterial modular PKS are colinear (per-module chemistry readable from domain content—hence the ClusterCAD database), whereas fungal PKS are iterative, so the program is not identifiable from the fixed domain set alone. A three-tier reconstruction analysis over all 326 fungal PKS entries in MIBiG 4.0 quantifies the ceiling: only **8/326** products are exactly reconstructible; a structural-similarity proxy is *unreliable* (swinging $8 \to 160$ with a single nuisance parameter, and admitting non-PKS false positives); and a tailoring-latent reformulation reconstructs **7/106** clusters exactly—but **64%** conditional on a producible core, localizing the bottleneck to PKS-core reconstructibility. A differential probe shows the per-step chemistry is learnable from abundant bacterial data and transfers to fungi perfectly with the active domains (1.00) but collapses to 0.57 from the fixed domain set—the iteration-grammar wall, which decomposes into a narrow highly-reducing control wall ($\sim$9%) and a broad, coverage-addressable aromatic-cyclization wall ($\sim$61%).

Finally we show the architecture is not PKS-specific: the same observability and verdict layers run *unchanged* over a non-ribosomal peptide (NRPS) grammar and a typed PKS–NRPS hybrid bridge—only the program generator is family-specific. All tools, data, and analyses are open and reproducible from single commands.

1 Introduction

Many fungal drugs—the cholesterol-lowering agent lovastatin, the antifungal griseofulvin—are polyketides: large carbon scaffolds assembled by a multidomain enzymatic factory, the polyketide synthase (PKS), encoded by a contiguous biosynthetic gene cluster (BGC). Genome sequencing has revealed that filamentous fungi carry vast numbers of such clusters, most of them *cryptic*: no metabolite has ever been linked to them. They are widely regarded as one of the most promising untapped reservoirs of new bioactive chemistry.

Turning a cryptic fungal BGC into a candidate molecule requires answering: *what* would the cluster make? Recent machine learning has attacked this with limited success. Riedling et al. (2024) trained classical classifiers to predict the bioactivity class of fungal secondary metabolites from BGC features; trained on 314 fungal BGCs they reached balanced accuracies of 51–68%, and adding 1,003 bacterial BGCs moved this only to 56–68%. That second result is our point of departure: if the bottleneck were a shortage of fungal examples, a fourfold increase in (cross-domain) training data should help substantially. It did not.

The deeper issue is the framing. *Natural-product discovery should not be posed as direct* BGC $\to$*structure regression, but as verifier-grounded inference over latent biosynthetic programs constrained by genome-derived grammars and physical observables.* A BGC is not a molecular blueprint but a constrained program space: the product is the image $y = \mathrm{Exec}_\Theta(z; G)$ of a latent program z under a biochemical operator library Θ and a domain-derived grammar G, and mining is inference over z from the observables in hand. We realize this as a **grammar-agnostic biosynthetic inverse compiler** (Figure 1): a grammar-specific executor enumerates legal programs, a shared observability layer collapses them with mass and MS/MS, and a shared verifier returns a three-state reconstructibility verdict. We do *not* claim a complete genome-to-structure predictor; we contribute the architecture, an empirical reconstructibility analysis of fungal PKS that explains the field's ceiling, and cross-grammar demonstrations. Our contributions are:

1. A **grammar-agnostic inverse compiler** over biosynthetic program grammars: a sound deterministic executor and beam verifier returning the exact consistent program set $Z^\star(y)$ behind one observable-constrained interface (Sections 3–4).
2. A **deterministic reconstructibility analysis** of fungal iterative PKS over MIBiG 4.0 (Section 6): exact-match 8/326, a core-match *sensitivity* showing structural-similarity proxies are unreliable, and a tailoring-latent reformulation (7/106; 64% conditional on a producible core), with a curated literature benchmark (Section 5) and a differential probe of the iteration-grammar wall (Section 9).
3. A **shared metabolomics observability layer** (Section 8): an adduct-aware, ppm-tolerant approximate verifier $Z^\star_\epsilon$, a within-grammar formula-ambiguity report, a tolerance-scored MS/MS matcher, and a minimum-sufficient-observables planner.
4. **Cross-grammar demonstrations** on PKS, NRPS, and a typed PKS–NRPS hybrid bridge (Section 10)—the same observability and verdict layers, only the program generator changing.
5. A **three-state reconstructibility verdict** (VERIFIED / UNDER-OBSERVED / OUT-OF-GRAMMAR) with a next-observable planner, resting on a failure-mode taxonomy (operator coverage, hidden iterative control, tailoring under-constraint, non-local structure) that replaces a single accuracy score.

All numbers in this paper are produced by the open code base by a single documented command and are traceable to a git commit; Appendix A lists the commands.

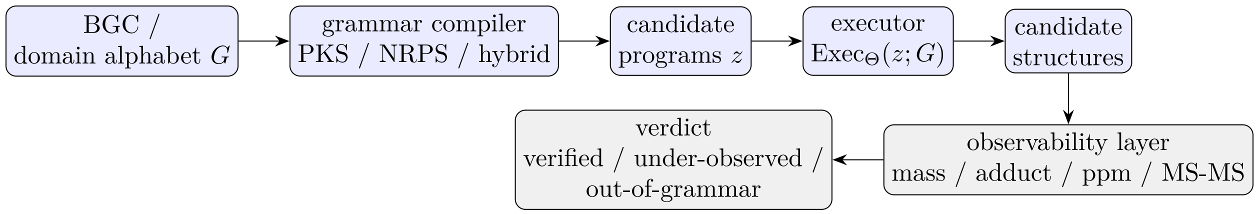

Figure 1: The grammar-agnostic inverse compiler. Family-specific grammars (blue) compile a cluster's domain alphabet into candidate programs that a sound executor renders to structures; a shared observability layer (grey) collapses the candidate set with mass and MS/MS, and a shared verifier returns a three-state reconstructibility verdict. Only the grammar and executor are family-specific; the observability and verdict layers are reused across PKS, NRPS, and hybrid grammars.

2 Background and the grammar-mismatch hypothesis

Modular vs. iterative assembly lines. Bacterial type I modular PKS—the archetype being the 6-deoxyerythronolide B synthase—are *colinear*. Each elongation module carries its own ketosynthase (KS) and acyltransferase (AT) for chain extension and an optional reductive cassette of ketoreductase (KR), dehydratase (DH) and enoylreductase (ER). The final product can be read off the linear order of modules. This clean structure-to-function map is why rule-based detectors such as antiSMASH (Blin et al., 2023) and PRISM (Skinnider et al., 2020) and the design database ClusterCAD (Eng et al., 2018; ClusterCAD 2.0, 2023) perform well on bacteria.

Fungal PKS are overwhelmingly *iterative*: a single set of catalytic domains is reused across N cycles, and which reductive steps fire on which cycle is governed by a program that no current tool decodes. The highly-reducing (HR), non-reducing (NR) and partially-reducing (PR) subclasses behave like distinct programming languages. The lovastatin nonaketide synthase LovB runs eight cycles with a different reduction state each time, and its own ER domain is degenerate—ER activity is supplied *in trans* by the standalone partner LovC (Ma et al., 2009). In NR-PKS a dedicated product-template (PT) domain dictates the regioselectivity of aromatic-ring cyclization (Crawford et al., 2009).

The hypothesis, sharpened to reconstructibility. We hypothesize that the 51–68% ceiling reflects a control-flow mismatch rather than a sample-size limit. We state this as a *reconstructibility* (identifiability) property, not undecidability: throughout, "computable" is shorthand for whether the per-step core is recoverable from the BGC-visible domain content under our sound executor grammar. In colinear assembly lines the per-step core *is a function of domain content* and can therefore be tabulated (ClusterCAD); in iterative assembly lines the same domain set must perform different chemistry on different cycles, so the per-step core is *not* a function of domain content and the end-product-to-program map is many-to-one and unobserved. A model trained on colinear data learns a control flow that is not merely uninformative but incorrect for fungi—the near-flat cross-domain result of Riedling et al. (2024) is exactly the prediction this hypothesis makes. The rest of this paper turns the hypothesis into measurements.

3 A verifier-grounded executor

We represent the product of an iterative synthase as the output of a discrete latent *program* executed by a sound, deterministic chemical executor (Figure 2). A program is an ordered list of catalytic

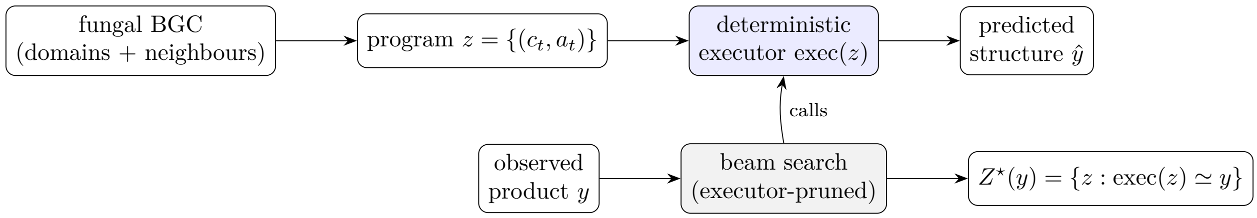

Figure 2: The executor (top) renders a program to a structure; the verifier (bottom) inverts it by beam-searching programs, executing each candidate and pruning any whose partial product cannot reach y, returning the exact consistent set $Z^\star(y)$.

cycles,

$$z = \big\{(c_t, a_t)\big\}_{t=1}^{N}, \tag{1}$$

where c_t marks a chain-elongation event and a_t is an action over the reductive cascade and tailoring chemistry. The cascade is strictly ordered—ER presupposes DH presupposes KR—giving the four reachable β-states (keto, hydroxyl, enoyl, methylene) plus an optional α-methylation (C-MeT) and a terminal chain release. The executor renders a program into a molecule, $\hat{y} = \mathrm{exec}(z)$.

Deterministic core, not weights. Each operator is a fixed reaction encoded as a reaction SMARTS and applied with RDKit (Landrum, 2023): a decarboxylative Claisen condensation appends a C_2 ketide unit; KR reduces the β-ketothioester to the β-hydroxy; DH eliminates to the enoyl; ER reduces to the methylene; C-MeT α-methylates; the thioesterase/cyclase releases the chain. The growing chain is modelled as an ACP-tethered thioester whose carrier is a single sentinel atom, so the unique active thioester anchors every operator unambiguously. The core is *validated*, not trained: we confirm each operator by exact mass bookkeeping (extension $+C_2H_2O$, KR $+H_2$, DH $-H_2O$, ER $+H_2$, C-MeT $+CH_2$) and by reconstructing known polyketides. Single-mode cyclization—lactonization, the small-aromatic aldol (orsellinic, 6-MSA) and the mellein dihydroisocoumarin—is implemented; the five-class PT-domain regiochemistry of large NR-PKS and the lovastatin Diels–Alder are deliberately scoped out and flagged. The executor is a pure, deterministic function validated within the project's regression suite (116 tests).

4 The verifier and identifiability

Because the executor is sound and deterministic and the program space is small (N typically 4–10, roughly eight admissible actions per cycle), we recover programs by enumeration rather than amortized inference. Given a target y and an operator set, a beam search emits programs, executes each (partial) program, and prunes any whose partial product cannot reach y. We accept a program into the *verified consistent set*

$$Z^\star(y) = \{\, z : \mathrm{exec}(z) \simeq y \,\}, \tag{2}$$

where $\simeq$ is graph isomorphism up to canonicalization. Two prunes make the search cheap: a carbon-count bound and an *analytic molecular-formula* prefilter that computes a program's released-acid formula from validated per-cycle deltas and only executes formula-consistent candidates (e.g. this takes the mellein search from 1,792 to 93 executor calls).

Validation and identifiability. On the curated benchmark the known program is contained in $Z^\star(y)$ for all systems, and the search is deterministic and fast (mean 0.28 s/target). Observed

Table 1: Curated benchmark (excerpt). Reduction states: K keto, H hydroxyl, E enoyl, M methylene. Gate-tier entries round-trip exactly through the executor.

System	Subclass	Program (starter + cycles)	Tier
Triacetic acid lactone	NR	acetyl + K,K → lactone	gate (sanity)
Orsellinic acid	NR	acetyl + K,K,K → aldol	gate
6-Methylsalicylic acid	PR	acetyl + K,H,K → aldol	gate
Mellein	PR	acetyl + H,K,H,K → dihydroisocoumarin	gate
6-Hydroxymellein	PR	acetyl + H,K,K,K → dihydroisocoumarin	gate
2-Methylbutyric acid (LovF)	HR	acetyl + M_{+Me}	gate
LovB nonaketide	HR	acetyl + 8 cycles (+DA)	hard / Phase-2
Tenellin backbone (TenS)	HR	acetyl + 4 cycles (+NRPS)	Phase-2 trace

degeneracy on the curated set is $|Z^\star(y)| = 1$: programs are identifiable at these chain lengths. Genuine degeneracy *is* detectable when the operator set permits an alternative route—e.g. hexanoic acid is reached as acetyl + $2\times$ER or butyryl + $1\times$ER, giving $|Z^\star(y)| = 2$—so identifiability is governed by the operator-set constraints, as the formulation predicts.

4.1 Guarantees

All guarantees are stated relative to the operator library Θ, the (domain-derived) grammar G, the observable set O, and—for the tolerant mass observable (Section 8)—the mass tolerance ϵ and the adduct model. For $Z^\star_\Theta(y; G, O) = \{\, z \in Z_\Theta(G, O) : \mathrm{Exec}_\Theta(z; G) \simeq y \,\}$ the search is *sound*—every returned z executes to a structure matching y under $\simeq$—and *relatively complete*: every legal program in the enumerated grammar that matches y is returned, up to the beam bound. It is deliberately *not* biologically complete: if nature uses an operator outside Θ, an empty $Z^\star$ is correct relative to Θ even though the molecule is producible *in vivo*. An empty consistent set is therefore a typed *certificate of failure mode*—missing operator, hidden control, tailoring under-constraint, or non-local pathway—not a claim of biological impossibility; the approximate verifier $Z^\star_\epsilon$ inherits the same relativity, now also to ϵ and the allowed adducts.

5 A curated benchmark of fungal PKS programs

We hand-curate twelve literature-sourced fungal polyketide programs (Table 1), each a machine-readable specification of starter, per-cycle reduction/methylation, and release, with a literature citation, a provenance tag (*independent* ground truth vs. *pending* an authoritative structure), and a confidence flag. Six gate-tier entries with independent ground truth round-trip exactly through the executor; sanity entries cover the reductive extremes; harder systems (the LovB nonaketide; the TenS PKS–NRPS backbone) are retained as trace-level assets.

A finding from curation is worth stating: the executor caught a mass-balance inconsistency in a literature-pass program for mellein. The single-ketoreduction program reconstructs 6-hydroxymellein ($C_{10}H_{10}O_4$), not mellein ($C_{10}H_{10}O_3$); mellein requires a *second* ketoreduction. The verifier earns its keep by making such errors impossible to overlook.

6 Three-tier reconstruction over MIBiG 4.0

Dataset. From MIBiG 4.0 (3,013 BGCs) we keep entries whose biosynthetic class includes PKS (1,238) that carry a compound with a resolvable structure (1,126) and whose producer is fungal

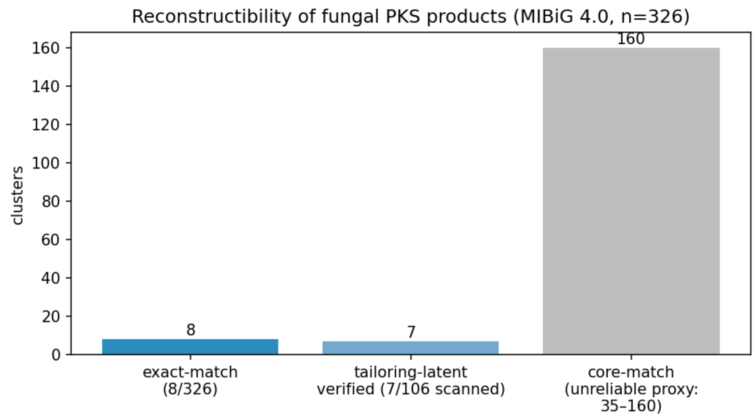

Figure 3: Reconstructibility of fungal PKS products (MIBiG 4.0, $n = 326$). Exact-match reconstructs 8; tailoring-latent verifies 7 of the 106 gene-budgeted clusters scanned; core-match (right, grey) is a permissive upper bound shown for contrast, not a trainable set.

by NCBI taxonomy lineage, yielding 326 **usable fungal PKS→structure pairs**. HR/NR/PR subclass is *unknown* for 304/326 because MIBiG does not module-annotate fungal iterative PKS (it reads colinear bacterial domains, not iterative ones)—itself a symptom of the grammar mismatch. We then ask, at three increasingly permissive levels, how many products our verifier-grounded tools can reconstruct (Figure 3).

6.1 Tier 1 — exact match: 8/326

Searching for a PKS program whose executor image equals the product exactly yields 8/326 reconstructible products (a lower bound under the *implemented* executor grammar—an artefact of operator coverage, not an absolute property of fungal PKS biology): small untailored aromatics and lactones (orsellinic acid, 6-MSA, the triacetic-acid-lactone family) plus a handful of reduced linears. This is the quantitative face of the computability claim: only the untailored, single-mode-cyclized fraction is directly reconstructible.

6.2 Tier 2 — core match is an unreliable proxy

One might relax exact match to a structural-similarity "core match": does a producible core skeleton embed in the product, absorbing additive tailoring? We implement this carefully (ring-aware carbon-skeleton subgraph, with a decoration-completeness criterion) and report it as a *cautionary sensitivity analysis* rather than a trainable set. The count is acutely sensitive to a single nuisance parameter—how many extra ring-closure bonds count as "tailoring"—swinging from 8 at exact match to 160 as the bound is loosened (Figure 4); a raw subgraph match admits 309/326. Worse, the carbon-only abstraction collapses heteroatom-ring non-PKS molecules (e.g. the alkaloid swainsonine) into carbon chains that a polyketide core "matches." **We conclude that structural-similarity proxies are unreliable for measuring reconstructibility here**, and do not use them to define the trainable set.

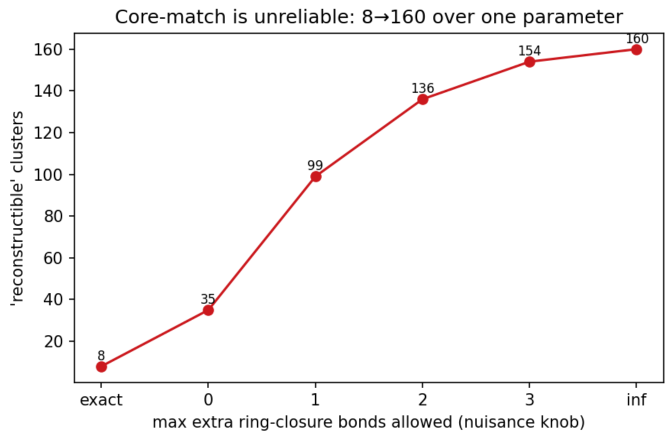

Figure 4: Core-match is unreliable: the "reconstructible" count swings from 8 (exact) to 160 as a single nuisance parameter (allowed extra ring-closure bonds) is varied.

6.3 Tier 3 — tailoring in the latent: 7/106, and the decisive diagnosis

The honest fix is to put tailoring *into* the latent: extend the program to $z = (z_{\text{pks}}, z_{\text{tail}})$, where the executor is the sound PKS executor followed by a tailoring-edit applier, so $\text{exec}(z)$ can reach the actual product and we score by exact match again. The objection—tailoring is unbounded—is defeated by two *observed* budgets: a **gene budget** (the cluster's co-located tailoring enzymes, typed from MIBiG annotations and GenBank CDS descriptions, bound which edit families may fire) and a **Δ-formula budget** ($\Delta = \text{formula}(y) - \text{formula}(\text{core})$ pins the edit-type multiset exactly). With ~ 12 deterministic edit types (hydroxylation, O/C-methylation, halogenation, prenylation, glycosylation, oxidative cyclization / decarboxylation, reduction/desaturation) and a parsimony prior, the joint search is tight and verifier-pruned.

Over the 106 gene-budgeted, size-tractable clusters, 7 **are exactly verified**, with a *near-unique* $|Z^\star(y)|$ distribution $\{1{:}6,\ 3{:}1\}$ (Figure 5). Six are untailored cores; one (stipitatic acid) is genuinely tailoring-unlocked. The decisive result is the diagnosis of the 99 non-verified clusters: 95 **are PKS-core-unreachable** (no producible backbone) and only 4 are tailoring gaps. Hence *conditional on a producible PKS core (*11/106*), the joint search verifies* 7*—*64%*—near-uniquely.* The tailoring layer works where it can apply; the wall is PKS-core computability (the unimplemented PT-aromatic classes and complex scaffolds), not tailoring.

6.4 The genuine ceiling

A subset of products is unreconstructible *in principle*, for three verifier-diagnosable reasons: (i) *oxidative skeletal rearrangement*—aflatoxins and sterigmatocystin (the norsolorinic-acid anthraquinone is rebuilt by Baeyer–Villiger ring expansion and bisfuran formation) and patulin (ring cleavage); (ii) *not a single-chain type-I PKS product*—the alkaloid swainsonine, the flavonoids naringenin/chlorflavonin; and (iii) *dimers and depsides*—aurofusarin, phomoxanthone A, the bicoumarin kotanin, lichen depsides. The exact membership is metric-dependent ($\sim$17 in the documented analysis), but the three classes are robust and are themselves a clean empirical statement about what "BGC→structure" can mean for fungi.

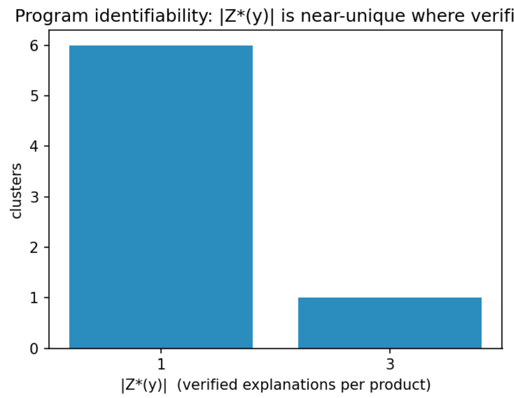

Figure 5: Explanation multiplicity for the seven tailoring-latent-verified clusters: six are uniquely explained and one has three consistent explanations. Where the joint search succeeds it is near-unique—the regime in which a verifier-constrained objective is well posed.

Where coverage should extend. A structural classification of the 318 non-exact-reconstructible products locates the coverage frontier empirically. The unreachable set is dominated by small monocyclic aromatics ($\sim$100) and small aliphatics/lactones ($\sim$58), together with N-containing PKS–NRPS hybrids and non-PKS skeletons ($\sim$98); the polycyclic anthrone/anthraquinone/xanthone class often invoked as the cyclization frontier is only ~ 16 clusters, mostly large ($C_{18–30}$) and largely coincident with the in-principle rearrangement ceiling above (sterigmatocystin, the cercosporin/elsinochrome perylenequinones). Extending the small-aromatic cyclization repertoire and modelling the PKS portion of hybrids would thus unlock substantially more than the higher PT-aromatic folds.

7 The conceptual result: grammar mismatch is the data deficit

The three tiers cohere into one statement. In colinear bacterial PKS the per-module core is a function of domain content, so it can be tabulated—ClusterCAD is the existence proof. In iterative fungal PKS the *same* domains must do different chemistry on different cycles, so the per-step core is not a function of the fixed domain set alone, so no fungal analogue of ClusterCAD can be built in the same per-module tabular sense without further observables; the end-product→program map is many-to-one and the per-cycle intermediates are unobserved. This is why augmenting scarce fungal data with abundant bacterial data does not help: the bacterial data teaches a control flow that is structurally wrong for fungi. The exact-match 8/326, the 304/326 unknown subclasses, and the 95/99 core-unreachable diagnosis are three faces of the same fact.

Two orthogonal walls, not one. The ceiling is not uniform, and conflating its parts overstates the computability claim. A structural subclass proxy[1] (aromatic-carbon fraction and saturation) splits the 326 products into three regimes. Non- and partially-reducing aromatics ($\sim$61%) have an architecturally fixed reduction program—a non-reducing synthase makes an all-keto chain—so their

[1] Per product: $N > 0$ heavy atoms ⇒ hybrid/non-PKS; else aromatic-carbon fraction $f_{\mathrm{ar}} \geq 0.4$ ⇒ NR/PR aromatic; $f_{\mathrm{ar}} \leq 0.1$ with H:C ≥ 1.4 ⇒ HR-reduced; otherwise PR/mixed. On the twelve curated systems this recovers the reduced-vs-aromatic axis for all four highly-reducing entries; it does *not* separate NR from PR (both aromatic), which the two-walls split does not require.

only hard step is cyclization regiochemistry, which is deterministic and *incrementally addressable by executor coverage*: generalising the resorcylic-aldol closure to an arbitrary starter alkyl, for instance, lifts exact reconstruction from 8 to 9/326 (olivetolic acid) with no loss of soundness. Highly-reducing polyketides ($\sim$9%) cyclize trivially but carry the genuine computability wall—the per-cycle reduction program, i.e. the 100%$\rightarrow$57% gap, shown above to be neither positional nor bacteria-transferable. The remaining $\sim$30% are N-containing PKS–NRPS hybrids or non-PKS skeletons, a scope boundary rather than a wall: the achiral C/H/O executor tracks the polyketide backbone up to the PKS–NRPS handoff (the linear checkpoint) but does not model the amino-acid condensation and release, so the PKS-core portion of a hybrid such as the tenellin synthase TenS stays computable even when its tetramic-acid product does not. Hence "fungal BGC→structure is uncomputable" is too strong: the computability wall is the narrow highly-reducing slice, while the dominant aromatic fraction is a tractable coverage problem.

8 From inverse compiler to a genome-mining engine

The same sound executor, run *forward* as a generator rather than inverted against a known product, is an operational genome-mining tool. Because the executor is a *sound generator*, a cluster's catalytic alphabet bounds a small enumerable set of producible cores—tiny when reduction is architecturally fixed, combinatorial for highly-reducing programs (a $\sim 13\times$ candidate-set ratio in our curated test)—which can be ranked and matched against metabolomics, recasting the reconstructibility ceiling as a constrained hypothesis generator. Genome mining is then inference over the latent program z from whatever observables are in hand, and the residual candidate set—which we measure operationally by its size, not by a calibrated entropy—falls as orthogonal observables are added (Table 2).

Table 2: The engine's input/output contract: each observable and what it constrains. Rows distinguish retrospective verification (known product) from prospective mining (unknown product).

Observable	Constrains	If absent / noisy
domain alphabet	legal operators (grammar)	broad candidate set
domain order / subclass	macro-grammar, release mode	wrong grammar
accurate mass (formula)	reduction / edit budget	formula-isomer ambiguity
MS/MS fragments	skeleton / isomer identity	residual isomers unresolved
co-clustered tailoring genes	admissible edit families	over/under-constrained tailoring
known product (benchmark only)	retrospective verification	prospective mining mode

The observable ladder. The genome alone leaves the program highly uncertain (the 0.57 per-cycle transfer); the domain alphabet bounds the legal operators; the MS-measured molecular formula pins the reduction *count*. Conditioning the generator on the two observables a metabologenomics pipeline actually supplies—domain alphabet and mass—collapses our curated reachable set to a median of *one* candidate core (top-3 recall 9/10), so the per-cycle reduction combinatorics, and with them the 0.57 grammar wall, largely dissolve once alphabet and mass are combined. Mass alone is not always sufficient: for small cores within executor reach alphabet+mass is near-deterministic, but larger cores under a rich grammar retain formula-isomers, and these are *fragmentation*-distinguishable (a crude single-bond-cleavage fingerprint already gives all 112 formula-isomers of one C_{12} core distinct fragment sets), so predicted MS/MS resolves the residual to a single core—a domain→mass→fragmentation pipeline grounded throughout in the sound verifier. The binding constraint is then cyclization *coverage*, not computability.

A grammar-relative identifiability benchmark. A controlled benchmark over 30 sampled cores bears the ladder out: the median verified set collapses $1672 \to 219 \to 1$ across grammar $\to$ +mass $\to$ +MS/MS, and the true core is recovered 30/30 under the correct grammar and true mass but 0/30 under a deliberately wrong, non-reducing grammar—so the grammar and the mass each do real work, and the small sets are not an artefact of a tiny search space. The intermediate 219 is the complex-core regime in which mass narrows sharply but not uniquely and MS/MS supplies the uniqueness; the median-1 collapse above is the tight-alphabet small-aromatic regime. We release a reference implementation, `lpi.engine`: a single interface mapping a cluster's domain set and observables (accurate mass, MS/MS) to ranked candidate cores and a per-cluster reconstructibility verdict.

A first real-cluster case study. To test the engine against the outside world rather than only curated or simulated targets, we ran it *blind* on a real MIBiG cluster—the 6-methylsalicylic acid synthase (`BGC0001275`, producer *Glarea lozoyensis*). MIBiG annotates the cluster only as class "PKS" with no per-module domains—the very gap this paper documents—so the grammar input is the canonical 6-MSAS architecture (KS-AT-DH-KR-ACP), and the sole observable is the real monoisotopic mass (152.0473, PubChem CID 11279) supplied as the $[\mathrm{M}-\mathrm{H}]^-$ ion at 5 ppm; the product structure is withheld from inference. The domain alphabet alone admits 1,121 producible cores, the accurate mass collapses these to a single candidate, and the engine returns VERIFIED. Only on reveal do we score it: the unique candidate is exactly 6-methylsalicylic acid (rank 1). A real EI-MS spectrum (MassBank `MSBNK-Fac_Eng_Univ_Tokyo-JP008039`) corroborates the molecular ion at m/z 152; being EI rather than LC-MS/MS it serves as a mass sanity reference, not a fragmentation rung, and since the mass alone is already decisive we report the MS/MS rung as not applied rather than simulate it. This is one clean real-cluster case, not a benchmark.

9 A path forward: differential learning, demonstrated

If absolute prediction (BGC→structure) is fungal-data-starved, the *differential* question—given two modules and the difference in their domains, predict the difference in their intermediate structures—is abundantly supervised, because ClusterCAD provides thousands of bacterial modules with per-module intermediate structures. The per-step chemistry a domain performs transfers across bacteria and fungi (a ketoreduction is a ketoreduction); only the iteration grammar differs. A differential model learns the transferable half directly. We run the remainder of this section as a sequence of controlled diagnostics: each isolates a candidate locus for the hidden per-cycle signal—gene content, then chain position—and asks whether it carries the information a predictor would need, so the experiments *localize* the iteration-grammar wall rather than merely score a model.

Setup. We scrape ClusterCAD (183 clusters → 1,752 modules → 1,091 clean extension modules), read per-step labels *from structure* (reduction state and KR stereochemistry at the active thioester), and train a small frozen model with domain-content features and two heads (reduction state; KR stereo), leave-cluster-out, against the antiSMASH-style domain rule and a majority baseline.

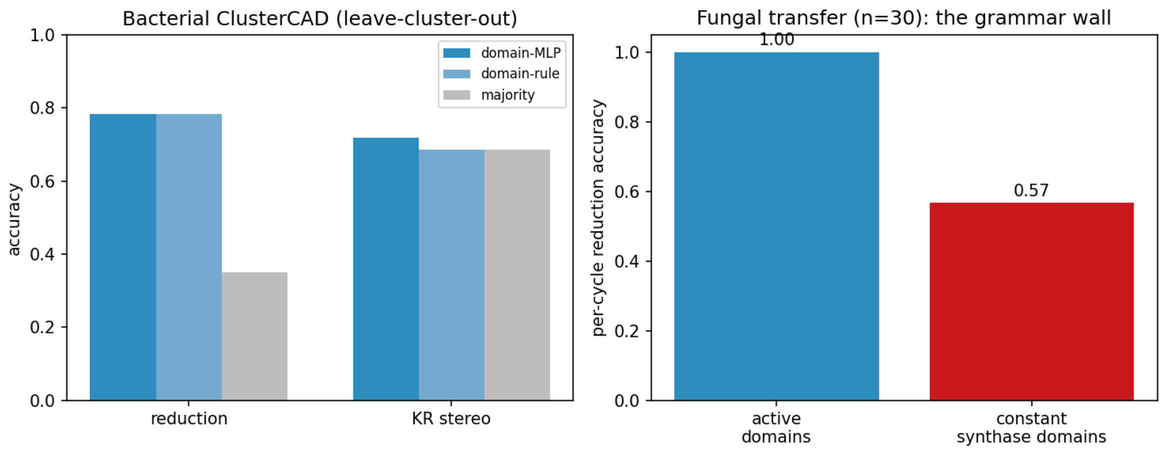

Figure 6: Differential PoC. Left: bacterial ClusterCAD (leave-cluster-out)—per-step reduction chemistry is learnable (domain-MLP $\approx$ domain-rule $\gg$ majority), while KR stereo is not domain-predictable. Right: fungal transfer—per-step chemistry transfers perfectly with active domains (1.00); given only the synthase's constant domain set it collapses to 0.57, the iteration-grammar wall.

Domains under-determine per-step chemistry—even in bacteria. Structure-derived reduction state agrees with the domain rule only 80.8% of the time over the 1,091 modules; the $\sim$19% gap is real domain-present-but-inactive events (a DH or KR that does not fire). This is the LovB-degeneracy phenomenon quantified at scale, in colinear bacteria.

Results (Figure 6). The reduction head reaches 0.78, equal to the domain rule and far above majority (0.35); DH is perfect, while KETO recall is only 0.30 (the inactive-domain ceiling). KR stereochemistry is *not* domain-predictable (0.72 $\approx$ 0.69 majority): it requires sequence motifs. The headline is a controlled diagnostic on the fungal side: we apply the bacteria-trained reduction head to 30 curated fungal cycles under two information conditions. Given the *active* domains, accuracy is 1.00 (per-step chemistry transfers perfectly); given only the synthase's *fixed* domain set it is 0.57. The contrast is the diagnostic: *that* 100% $\to$ 57% *gap is the iteration grammar*—the same genes must do different chemistry on different cycles, and gene content cannot say which.

The residual gap is not positional. A second diagnostic asks whether the missing per-cycle signal is the running intermediate's state, cheaply proxied by chain *position* (cycle index, distance from the terminus)—available at mining time and, unlike the active-domain oracle, not a leak of which domains fired. It is not. Conditioning the bacteria-trained head on position does not beat the constant-domain wall on the 30 fungal cycles (leave-one-synthase-out 0.53 versus the wall's 0.57; bacteria-trained zero-shot, 0.57). The cause is mechanistic: bacterial per-step reduction is *positionally flat*—a function of domain content, not of chain position, across all 1,091 modules—so there is no positional gate for a bacteria-trained model (positional or geometric) to transfer. Within-fungal positional structure does exist but is subclass-specific and does not generalise across synthases (partially-reducing cycles benefit; highly-reducing cycles are hurt). The residual gap therefore demands *sequence or structure* signal, not geometry imported from bacteria. This is a small-sample probe ($n = 30$ cycles, 9 synthases), but the bacterial-flatness result it rests on spans all 1,091 modules.

A planned head, blocked on data access. The natural next test—does *sequence* carry the chemistry signal that gene content lacks?—is implemented as a frozen ESM-2 (Lin et al., 2023) head over two bacterial targets (KR stereo; inactive-domain detection), but is *built but blocked*: the per-domain label→sequence join is exactly the bespoke module decomposition ClusterCAD performs, so the obstacle is a deliberate algorithmic boundary rather than a missing experiment. The head is coded and runnable the moment a per-domain-labelled sequence source exists. Details and the assembled label sets are in Appendix B.

10 Cross-grammar generalization: NRPS v0

The program-synthesis view of Section 7—a BGC is not a molecular blueprint but a constrained program space, $y = \mathrm{Exec}_\Theta(z; G)$—predicts that the engine should not be specific to polyketides: biosynthetic families differ in their program grammars and operator libraries Θ, but the *observability* layer (accurate mass, MS/MS) and the reconstructibility verdict are physical, not family-specific. We test this directly. Behind the same `infer_cluster`$(G, O) \to (Z^\star$, state, next observable) interface we implement a minimal non-ribosomal peptide (NRPS) grammar: a pure, deterministic peptide executor (amide condensation of a small monomer set; linear and head-to-tail macrocyclic release) with an exact residue-formula prefilter, presented through the identical `Grammar` protocol as the PKS generator (Table 3).

Table 3: One inference interface, two grammars. Verified candidate-set size after each observable rung, and the three-state verdict; the same mass / MS-MS / verdict machinery serves both families.

Cluster / product	Grammar	alphabet	+mass	+MS/MS	Verdict
orsellinic acid	PKS	2	1	1	verified
olivetolic acid	PKS	5012	87	1	verified
cyclo(Gly–Gly)	NRPS	100	1	1	verified
glycylglycine	NRPS	576	1	1	verified
cyclo(Phe–Tyr)	NRPS	100	1	1	verified
NRPS, mass withheld	NRPS	788	—	—	under-observed

Because the observability layer consumes only executed candidate structures, the same ppm-tolerant mass recovery, adduct model, MS/MS scorer, observable ladder, and three-state verdict operate unchanged across PKS and NRPS grammars; only the program generator is family-specific. The withheld-mass row is the engine in its operative mode: from the genome alone the peptide is *under-observed*, and the verdict names the observable—accurate mass—that would collapse the candidate set.

Scope. This is a proof of *framework* generality, not NRPS biochemical coverage (the v0 grammar's conservative scope is stated once in Section 12): the point is that a second grammar plugs into the same inference and verdict stack, the architecture being grammar-agnostic because the shared layer reads only candidate structures.

10.1 Compositional grammars and hybrid biosynthesis

Biosynthetic families also *compose*. A PKS–NRPS hybrid is modelled as a typed composition of the two grammars,

$$z_{\text{hybrid}} = z_{\text{PKS}} \circ h_{\text{handoff}} \circ z_{\text{NRPS}} \circ r_{\text{release}}, \tag{3}$$

where h_{handoff} is an explicit amide-condensation operator joining the polyketide *linear-acid checkpoint* (Section 3) to the peptide N-terminus. The hybrid grammar does not re-implement either family: it composes the PKS generator with the NRPS monomer set, and the same adduct/ppm-tolerant mass recovery, MS/MS scorer, observable ladder, and three-state verdict run over the composed candidates unchanged (a minimal hybrid, acetoacetyl-glycine, collapses genome→mass to a single *verified* core through the identical stack). Crucially, the fungal tetramic-acid Dieckmann release—the step that converts the TenS polyketide–tyrosine intermediate into pretenellin A—is declared as a *typed but unimplemented* operator: requesting it returns an explicit out-of-grammar verdict naming the missing release operator, rather than a silent failure or a wrong structure. The scope boundary the rest of this paper describes in prose is thus made machine-checkable. The framework now spans **PKS, NRPS, and a minimal PKS–NRPS hybrid bridge** behind one inference-and-verdict interface; grammars are family-specific and composable, while observability and the reconstructibility verdict are shared.

11 Related work

Riedling et al. (2024) establishes the 51–68% ceiling and the near-flat cross-domain result that motivates this work. Rule-based detectors and product predictors—antiSMASH/fungiSMASH (Blin et al., 2023), PRISM (Skinnider et al., 2020), DeepBGC (Hannigan et al., 2019), GECCO (Carroll et al., 2021), CLOCI (CLOCI, 2024)—detect clusters and, for bacteria, read off colinear products; none induce an iterative program. ClusterCAD (Eng et al., 2018; ClusterCAD 2.0, 2023) is the modular-PKS intermediate database we both consume (for the differential probe) and cite as the bacterial existence proof that colinear assembly lines are tabulable (the conceptual result of Section 7). MIBiG (MIBiG, 2023) is our pair source; BGC family methods such as BiG-SLiCE / BiG-FAM (Kautsar et al., 2021) define the within-family neighbourhoods relevant to differential pairs. Property/arrangement predictors such as CHAMOIS (CHAMOIS, 2025) infer ontology-level chemistry, not exact iterative programs. Methodologically we draw on execution-guided, neurosymbolic program synthesis (Ellis et al., 2021) and on protein language models (Lin et al., 2023); chemistry is handled with RDKit (Landrum, 2023). Our distinguishing stance is to *measure* exact reconstructibility and to ground every claim in a sound deterministic verifier.

12 Limitations

What this does *not* claim. We do not claim complete fungal structure prediction, complete NRPS or hybrid coverage, or validation on real untargeted-metabolomics spectra. We claim a grammar-agnostic, verifier-grounded inference architecture—an inverse compiler with a shared observability and three-state-verdict layer—and demonstrate it on PKS, NRPS, and a typed PKS–NRPS hybrid grammar. The NRPS and hybrid grammars are v0 demonstrations of architecture generality, not full family coverage. With that scope fixed once, the specific limitations are:

- **Partial executor coverage.** Single-mode cyclization plus a tetraketide-aromatic and a PT-naphthalene class (skeleton-correct; hydroxyl regiochemistry flagged) are implemented; the remaining PT-aromatic classes, the Diels–Alder, and macrolactonization are not. Thus the 95/99 core-unreachable count is partly executor coverage and partly true rearrangement.
- **Additive-only tailoring.** The $\sim$12 edit types are additive and the joint verifier is "exact-by-budget" (skeleton embedding + Δ-formula + gene/parsimony budget), threshold-free but not yet full atom-by-atom positional reconstruction (positions affect $|Z^{\star}(y)|$ multiplicity, not existence).

- **Gene budget over-counts** when a locus accession is a whole contig (non-cluster genes included); HMMER on cluster-scoped sequences would tighten it.
- **Subclass unknown for** 304/326; stereochemistry is not modelled (the executor is achiral by design); numbers are over MIBiG-deposited entries, biased toward well-studied (often heavily tailored) compounds.
- **No trained genome→structure model is claimed.** The differential reformulation demonstrates the transferable per-step chemistry; the iteration grammar (the 100% → 57% gap) remains open for *blind* prediction, though we show it is largely circumvented given an MS mass (Section 8).
- **The mining pipeline has one real-cluster case study, not yet a benchmark.** A blind real-cluster, real-mass run (6-MSA, Section 8) verifies the known product at rank 1; beyond it the alphabet+mass+MS/MS chain is shown as near-unique recovery on reachable cores and as fragment-fingerprint *separability* under a crude in-silico fragmenter. A multi-cluster benchmark, real LC-MS/MS validation (e.g. CFM-ID / GNPS), per-cluster domain alphabets from antiSMASH/fungiSMASH, and a learned candidate-ranking prior (data-limited at present) are the natural next steps. In clean form mass and MS/MS act as hard masks; in real metabolomics (adducts, isotopes, noisy or partial spectra) they become high-confidence constraints with explicit uncertainty; we *implement* a matching approximate verifier $Z^\star_\epsilon = \{\, z : \text{mass}(\text{Exec}(z)) \in [m-\epsilon, m+\epsilon],\ \text{MS/MS}(z) \geq \tau \,\}$ as an adduct-aware, ppm-tolerant, tolerance-scored layer shared across grammars (built via a mass-window formula prefilter so the tolerance is genuine, not an exact verifier with a cosmetic window) alongside the exact one—validation against real spectra (above) being the open step.

Table 4: Executor-coverage roadmap: operator extensions and the product classes they would unlock, turning "partial coverage" from a limitation into a prioritised programme.

Operator extension	Unlocks	Difficulty / risk
more small-aromatic closures	many NR/PR aromatics	low
PT-domain regiochemistry classes	larger aromatic folds	medium
macrolactonization	resorcylic-acid lactones, macrocycles	medium
PKS–NRPS handoff checkpoint	hybrid cores	high
oxidative skeletal rearrangement	aflatoxin / sterigmatocystin	high (ambiguous)
dimerization / depside coupling	non-local products	high (non-local)

13 Conclusion

This work does not add another black-box BGC→structure predictor; it builds a grammar-agnostic *inverse compiler* for biosynthetic programs—demonstrated on PKS, NRPS, and a typed PKS–NRPS hybrid—and uses its fungal-PKS instance to *measure* reconstructibility relative to a grammar. The fungal BGC-to-molecule ceiling is, we have argued and measured, a grammar problem masquerading as a data problem—and grammar mismatch is itself the data deficit, because iteration makes the per-step core non-identifiable from gene content alone. We quantified the reconstructible fraction (exactly 8/326; 7 of the 106 gene-budgeted clusters under a tailoring-latent model, but 64% conditional on a producible core), showed that structural-similarity proxies are unreliable for this question—similarity is not a reconstructibility metric, since reconstructibility is the existence of a legal program, not skeleton overlap—and located the wall precisely at PKS-core reconstructibility.

The path forward is differential: per-step chemistry is learnable from abundant bacterial data and transfers to fungi perfectly when the active domains are known—and the residual $100\% \to 57\%$ gap is a clean, quantitative target for future work on inducing the iteration program itself—and a *narrow* target: the ceiling decomposes into a small highly-reducing control wall and a broad, coverage-addressable aromatic-cyclization wall, and the same sound executor serves as a constrained candidate generator, turning a negative result into a usable mining tool. The tools, benchmark, and analyses are open and reproducible.

Outlook. Read through the program-synthesis lens of the introduction ($y = \mathrm{Exec}_\Theta(z; G)$), the failure modes form a reconstructibility taxonomy—operator coverage, hidden per-cycle control, tailoring under-constraint, and non-local pathway structure—and the field's single accuracy score aggregates distinct phenomena this decomposition separates. Deepening the family grammars (NRPS substrate specificity, the tetramic-acid release, a learned transition policy for the highly-reducing wall) and validating the observability layer on real spectra are the natural next steps. The architecture itself—grammar-specific generation under a shared observability and three-state verdict layer—is the contribution, with the reference implementation (`lpi.engine`) released alongside.

Reproducibility. All results are produced by single commands over pinned dependencies; see Appendix A.

A Reproducibility

Python 3.12, RDKit 2025.9.6; optional `torch`/`transformers` for the differential and ESM-2 components. Each result maps to one command: `make data` (the 326 pairs), `make roundtrip` (executor gate), `make search` (the verifier and $Z^\star(y)$), `make reachability` (exact-match 8/326), `make corematch` (the core-match sensitivity), `make phase0c` (tailoring-latent 7/106), `make clustercad` and `make differential` (the differential probe), `make crossgrammar`, `make observability`, and `make realdemo` (the cross-grammar PKS/NRPS/hybrid demo, the candidate-collapse observability benchmark, and the blind real-cluster 6-MSA mining demo), `make figures` (Figures 3–6), and `make test` (116 tests). The six tagged milestones (`lpi-v0.2-cross-grammar` through `lpi-v0.6-real-mining`), per-result numbers, scripts, and commits are tabulated in the repository `README` and `PAPER_READINESS.md`.

B The ESM-2 sequence head: built, blocked on data access

The differential probe (Section 9) shows the residual per-cycle signal is neither in gene content nor in chain position, leaving *sequence or structure* as its locus. The natural next test—does sequence carry the chemistry signal that gene content lacks?—is implemented as a frozen small ESM-2 (Lin et al., 2023) embedding with a tiny head, with two bacterial targets (KR stereo; inactive-domain detection) and the labelled set assembled (1,039 KR-bearing modules, 77 present-but-inactive KRs, 331 stereocentres). We report it honestly as *built but blocked*: ClusterCAD's per-domain sequence endpoint returned valid data on a first call and then reliably failed (HTTP 404) for bulk retrieval. We further attempted to reconstruct the per-domain sequences from primary data—MIBiG locus accessions → NCBI protein translations → a Pfam ketoreductase profile (PF08659) via HMMER—but the label→sequence join does not come clean: the structure-derived ClusterCAD labels count only confirmed C_2/C_3 extension modules, whereas the profile detects *every* ketoreductase domain (loading modules, standalone tailoring short-chain dehydrogenases, contig neighbours), reconciling for only a handful of clusters without reimplementing ClusterCAD's module parser. This is a deliberate algorithmic boundary rather than a missing experiment: the per-domain label→sequence link *is* the bespoke module decomposition ClusterCAD performs, and rebuilding it from raw genome sequence is out of scope here. The head is specified, coded, and runnable the moment a per-domain-labelled sequence source exists (a ClusterCAD export, or domain calls reconciled to ClusterCAD's module boundaries).

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